

Vishnu Bhargav Kotte

Fremont, CA, US

+1 (341) 252 9107 | kottevishnubhargav@gmail.com | [LinkedIn](#)

SUMMARY

AI Systems Engineer with 4+ years of experience building scalable backend systems, data pipelines, and LLM-powered applications across healthcare and enterprise environments. Experienced in prompt engineering, retrieval-based AI systems, evaluation workflows, and production-grade Python development. Strong background in backend engineering, ETL pipelines, API integration, and building reliable AI-native applications.

TECHNICAL SKILLS

- **AI & Large Language Models:** Large Language Models (OpenAI, Claude APIs), Prompt Engineering, Structured Prompt Design, Prompt Orchestration, LLM Workflow Automation, LLM Pipeline Design, Evaluation Pipelines, Embeddings, Semantic Retrieval, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Chroma, Pinecone), Structured Output Validation, NLP (Text Classification, Sentiment Analysis)
- **Machine Learning & Data Science:** Scikit-learn, Pandas, NumPy, Feature Engineering, Data Preprocessing, Model Training & Validation, Regression & Classification Models, Exploratory Data Analysis (EDA)
- **Programming:** Python (Advanced), Java, C/C++, SQL, PL/SQL, Data Structures & Algorithms, Object-Oriented Programming (OOP)
- **Backend & APIs:** Python Backend Development, REST APIs, FastAPI, API Integration, SQL Automation, Database Design, Stored Procedures, Triggers, JSON Data Processing
- **Cloud & Infrastructure:** AWS (EC2, IAM, VPC), Docker, Git & GitHub, CI/CD, HTTPS Deployment, Linux Environment
- **Databases:** Oracle PL/SQL, Relational Databases (RDBMS), Query Optimization, Indexing Strategies
- **Web Technologies (Working Knowledge):** React, TypeScript (Basic), HTML5, CSS3, Tailwind CSS

EXPERIENCE

PathAI, Remote, USA

AI/ML Engineer (LLM Systems)

January 2025 – December 2025

- Designed and implemented LLM-powered workflows for clinical and healthcare text interpretation, integrating structured prompts and domain-specific context to improve reasoning consistency.
- Designed modular prompt orchestration workflows with version-controlled templates to improve consistency across LLM-driven use cases.
- Developed Python-based evaluation pipelines to benchmark LLM outputs using qualitative review criteria and structured validation checks for factual consistency and formatting reliability.
- Built retrieval-based context grounding pipelines using embeddings and semantic search to improve LLM factual accuracy and reduce hallucinations.
- Engineered preprocessing pipelines for large-scale clinical text datasets using Pandas and NumPy to support experimentation and inference workflows.
- Optimized LLM inference workflows by reducing token usage, improving response latency, and standardizing prompt-response formats for cost-efficient execution.
- Collaborated with cross-functional research and engineering teams to translate applied AI research concepts into scalable prototype systems.
- Documented experimentation results, prompt strategies, and evaluation methodologies to support reproducibility and continuous system improvement.

Cognizant Technology Solutions, Hyderabad, India

Associate

October 2022 - December 2023

- Supported high-volume enterprise billing and production systems for Texas-New Mexico Power (TNMP), ensuring operational reliability, SLA compliance, and production pipeline stability.
- Diagnosed and resolved production pipeline failures and transactional data inconsistencies in enterprise systems.
- Took part in on-call production support, handling client communications, addressing billing interruptions, resolving high-priority incidents and ensuring SLA compliance.
- Prepared and reviewed process documentation, SOPs and handover note to streamline shift transitions and team knowledge.
- Led quality checks, reviewed work done by junior team members and supported client-facing updates and reporting.

Programmer Analyst

August 2021 - September 2022

- Developed SQL/PLSQL automation workflows and supported enterprise billing systems within utility operations environments.
- Supported daily production tasks in the Banner CIS & EDI systems, ensuring timely execution of billing chains, data uploads and transactional workflows.

- Developed and enhanced SQL/PL-SQL scripts to automate repetitive billing and reporting tasks, reducing manual effort and minimizing human errors.
- Collaborated with cross-functional teams (billing, metering, customer data, IT ops) to meet client deliverables and resolve business-critical issues.
- Assisted new team members by facilitating knowledge transfer sessions and mentoring them on CIS process flow, data structures and escalation procedures.

Novartis, Remote, India

Data Engineer

July 2020 - June 2021

- Built and maintained Python-based ETL pipelines to process multi-source healthcare datasets (CSV, relational DB exports, API feeds), transforming raw data into analytics-ready tables for internal reporting teams.
- Designed SQL transformation workflows to aggregate patient-level and operational datasets, reducing manual data preparation time by ~40%.
- Implemented automated data validation scripts (null checks, schema validation, anomaly detection rules) to improve dataset reliability before downstream analytics and modelling.
- Developed reusable preprocessing modules in Python (Pandas, NumPy) for feature engineering and dataset normalization to support statistical and ML experiments.
- Optimized query performance for large datasets (100k+ records) by restructuring joins and indexing strategies, reducing execution time significantly.
- Built scheduled batch jobs to automate recurring reporting workflows, ensuring consistent weekly and monthly dataset refresh cycles.
- Documented end-to-end data pipeline architecture, including ingestion, transformation, validation, and output layers, improving reproducibility across teams.
- Collaborated with data scientists to refine dataset structures and feature schemas for modeling experiments and analytical dashboards.

PROJECTS

Retrieval-Augmented AI Knowledge Assistant | RAG, LLM Pipelines, FAISS, FastAPI, Python

- Built an end-to-end Retrieval-Augmented Generation (RAG) system that answers user queries using contextual document retrieval and LLM-powered response generation.
- Implemented document ingestion, preprocessing, chunking, embeddings generation, and semantic similarity search using FAISS for efficient retrieval workflows.
- Designed prompt grounding and context injection strategies to reduce hallucinations and improve response relevance and factual consistency.
- Developed Fast API-based backend endpoints for scalable query handling, document processing, and AI pipeline integration.
- Optimized retrieval depth, prompt payload size, and inference workflows to improve latency, response efficiency, and token cost management.

LLM-Powered Career Automation Platform | Python, Claude API, Prompt Engineering, JSON Schema Design, ReportLab, python-docx

- Architected and developed an LLM-powered career automation platform using Python and Claude API to generate context-aware resumes, professional emails, and job application materials.
- Designed modular prompt architectures and structured JSON-based data schemas to ensure deterministic prompt conditioning and consistent LLM output quality.
- Built automated document generation pipelines (python-docx, ReportLab) to produce ATS-optimized PDF/DOCX outputs with standardized formatting and validation checks.
- Reduced manual resume customization time by ~95% (from 2+ hours to under 5 minutes) through workflow automation, prompt engineering, and structured evaluation logic.

EDUCATION

University of the Pacific

Jan 2024 - Dec 2025

Master, Computer Science (GPA: 3.78)

Stockton, CA

Chaitanya Bharathi Institute of Technology

Aug 2017 - June 2021

Bachelors, Electronics and Communication Engineering (GPA: 3.4)

Telangana, India